Understanding Variational Inference

A concise, self-contained example for exploring research-paper concepts

Abstract

Probabilistic models often contain latent variables whose posterior distributions are difficult to compute. Variational inference turns posterior inference into an optimization problem by choosing a tractable family of approximate distributions. This note introduces the evidence lower bound, explains amortized inference, and highlights the assumptions behind the method.

1 The inference problem

Suppose we observe data x and posit an unobserved variable z. Bayes rule says that the posterior p(z | x) is proportional to the joint distribution p(x, z). Computing the normalization constant p(x) requires integrating over all possible values of z. For expressive models, that integral may be intractable.

Instead, we choose an approximate distribution q(z) from a family that is easier to evaluate and sample. The goal is to find a member of that family that is close to the true posterior. One common measure of closeness is the Kullback-Leibler (KL) divergence.

2 The evidence lower bound

We optimize the evidence lower bound (ELBO) rather than the marginal likelihood directly. The ELBO contains two terms: an expected log likelihood, which rewards explaining the data, and a KL divergence, which regularizes the approximate posterior toward the prior.

$$\mathrm{ELBO} = \mathbb{E}_{q(z|x)}\left[\log p(x|z)\right] - \mathrm{KL}\left(q(z|x) \,\|\, p(z)\right)$$

The term lower bound is literal: log p(x) equals the ELBO plus KL(q(z|x) || p(z|x)). Since KL divergence is nonnegative, the ELBO cannot exceed log p(x). Maximizing it indirectly moves q toward the posterior, subject to the chosen variational family.

3 Amortized variational inference

Optimizing a separate variational distribution for each observation can be expensive. Amortized variational inference uses a shared inference network with parameters phi. The network maps an observation x to the parameters of q_phi(z | x), so inference for a new observation becomes a forward pass.

We optimize the evidence lower bound using amortized variational inference. The approximate posterior is parameterized by an inference network. The generative model has parameters theta, while the inference network has parameters phi. Both can be trained jointly with stochastic gradients.

3.1 Why amortization helps

The cost of learning a shared network is paid during training. At test time, the learned network quickly predicts variational parameters for unseen observations. The tradeoff is an amortization gap: a shared predictor may give a less accurate approximation than optimizing local parameters separately for each data point.

4 A useful identity

A compact derivation begins with the definition of KL divergence between q(z | x) and the true posterior p(z | x). Substituting Bayes rule and rearranging terms gives the identity below.

$$\log p(x) = \mathrm{ELBO}(q) + \mathrm{KL}(\, q(z|x) \,\|\, p(z|x) \,)$$

This identity also reveals a limitation. The ELBO can be loose if the variational family cannot represent the posterior well. A high ELBO within a restrictive family does not prove that the model captures every structure in the observed data.

5 Reading the method critically

When reading a paper that uses variational inference, check which distribution is approximated, what family is used for q, whether gradients are unbiased or estimated, and how the reported objective relates to the true log likelihood.

For continuous latent variables, authors often use the reparameterization trick. A sample z from q_phi(z | x) is expressed as a deterministic function of parameters and a separate noise variable. This allows gradients to flow through the sampling operation under suitable assumptions.

Questions to ask

Where did the ELBO come from? Why is it a lower bound? What is the difference between a prior and an approximate posterior? Which terms are estimated with Monte Carlo samples? Does the paper provide enough detail to verify the claimed optimization objective?

Notes

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